Chapter 3

Rectifiers

Instructor: Le Hoa Nguyen, Ph.D.

Part 1

Diode Rectifiers
(AC to Uncontrolled DC)

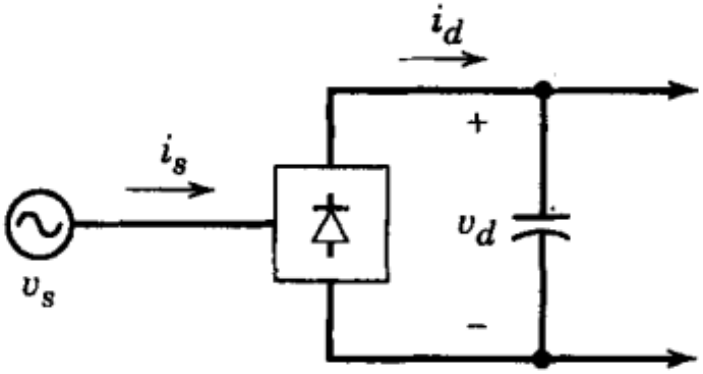
Line-frequency diode rectifier

Rectifier: An electrical device that converts alternating current (AC) to direct current (DC)

Line-frequency diode rectifier:

- Conversion: Line-frequency AC to uncontrolled DC
- Applications: power supplies, ac motor drive, variable speed DC drives, battery chargers,…
- Low cost, popular
- Large capacitor needed for ripple-free DC voltage
- Draw highly distorted current from utility

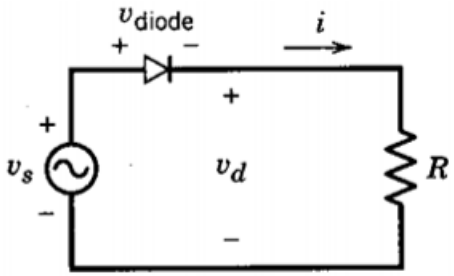
Note: Diodes are assumed to be ideal in the analysis of rectifiers



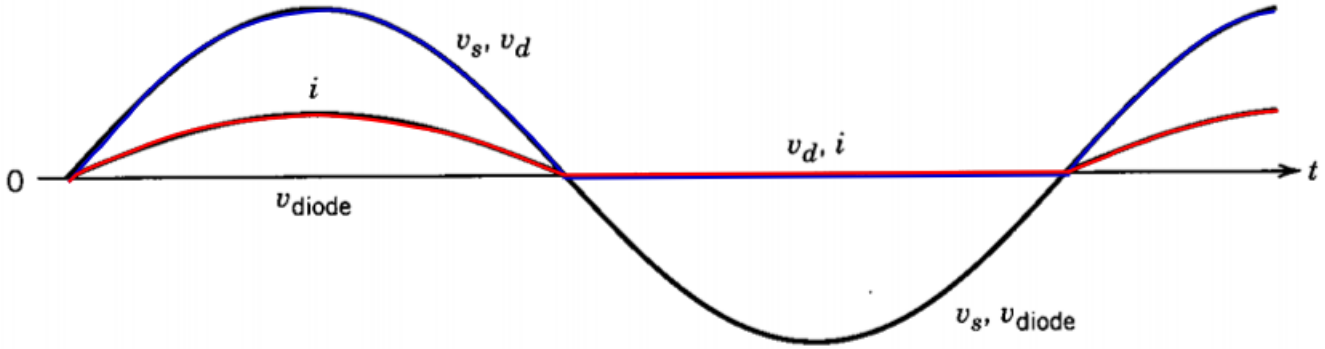
Single-phase diode rectifiers

Pure resitive load:

- Very simple, basic rectifier
- Diode conducts (forward-biased) for the positive half-cycle of the source
- Large ripple in v_d



(a)

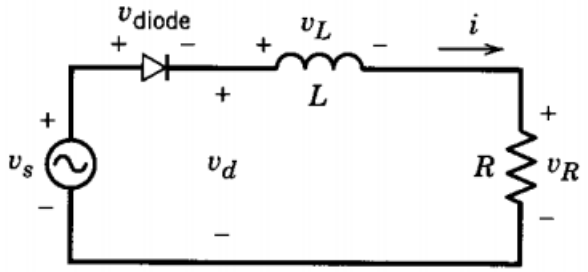


(b)

Single-phase diode rectifiers (half-wave rectifiers)

Inductive load:

- Industrial load typically contain inductance as well as resistance
- KVL when the diode is forward-biased

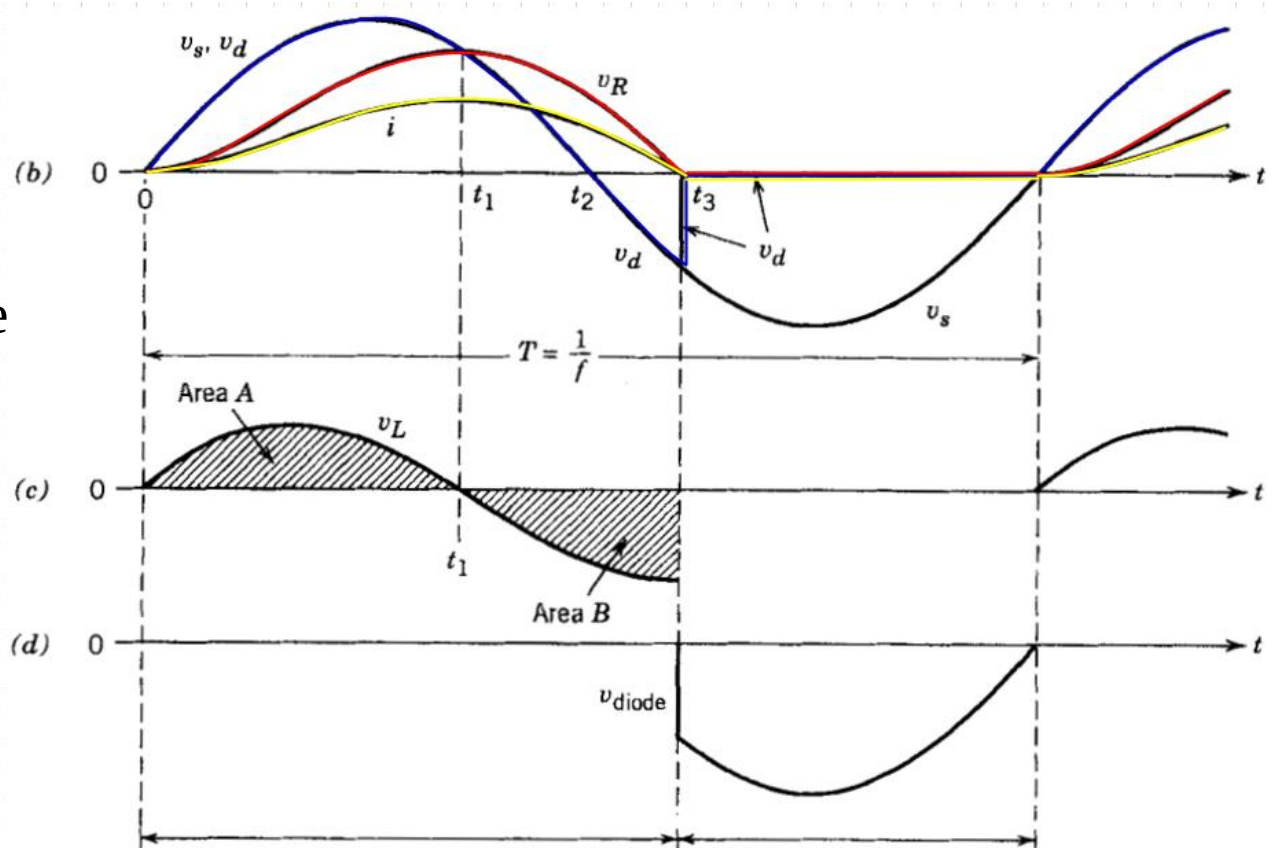


$$v_s = v_R + v_L$$

$$= Ri + L\frac{di}{dt}$$

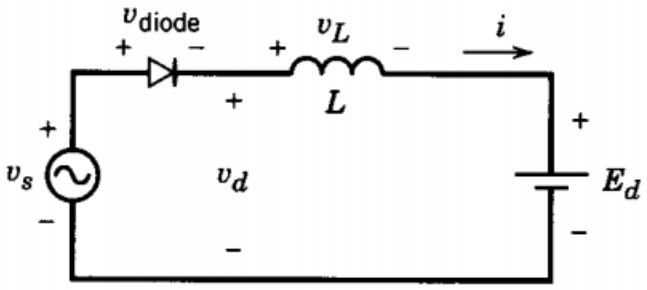
- Average voltage across L

$$V_{L,avr} = \frac{1}{T}\int_0^{t_3} v_L dt = 0$$



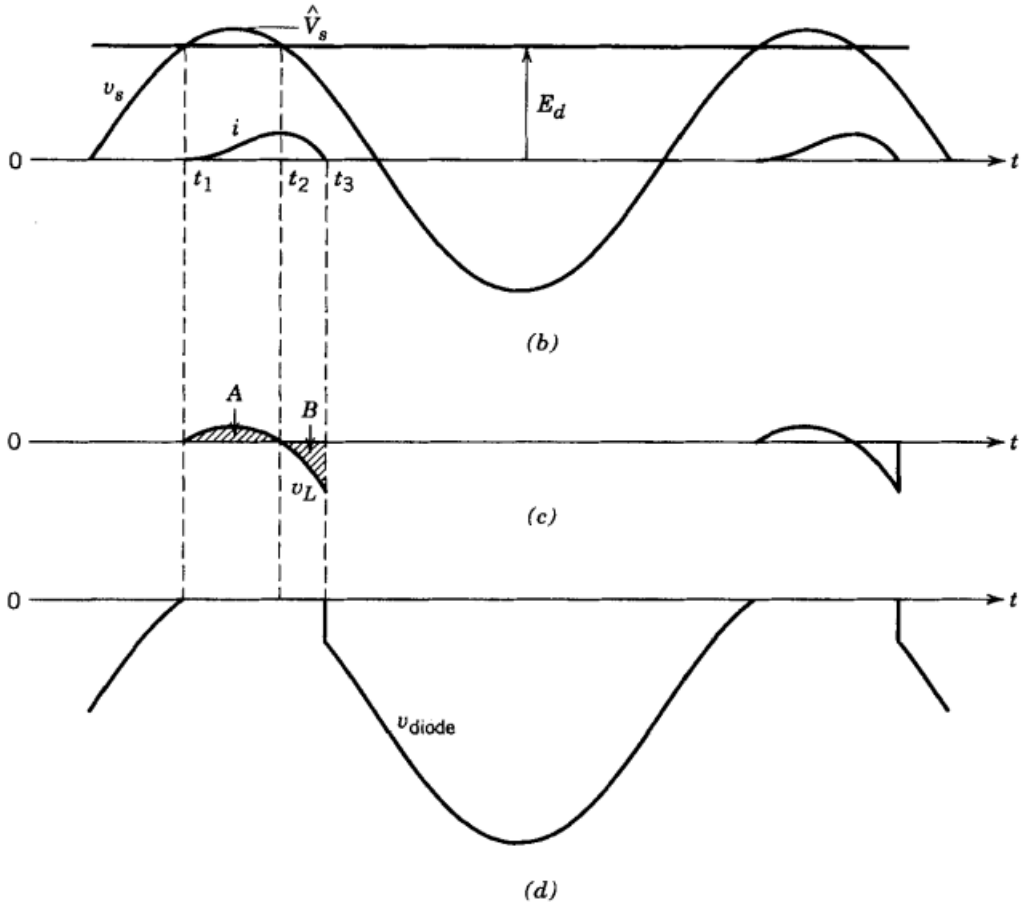
Single-phase diode rectifiers (half-wave rectifiers)

Load with internal DC voltage/large capacitor:



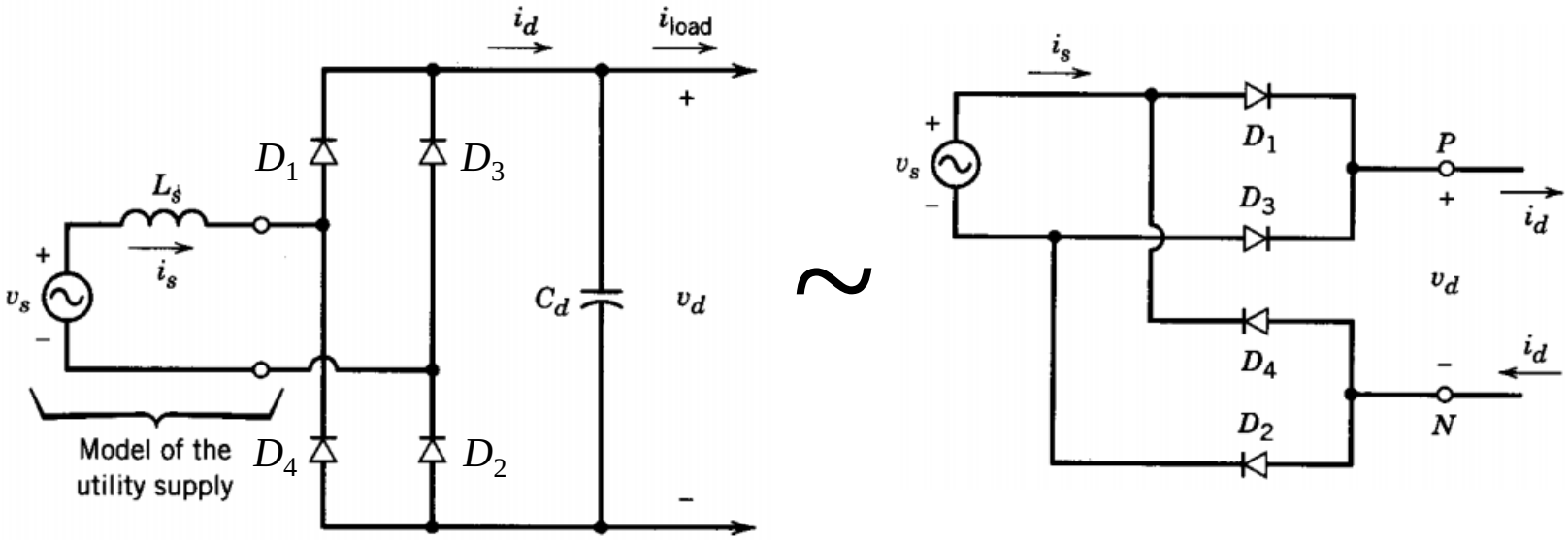
- The diode is forward-biased if $v_s > E_d$
- Average voltage across L

$$V_{L,avr} = \frac{1}{T}\int_{t_1}^{t_3} v_L dt = 0$$



Single-phase diode bridge rectifiers (full-wave rectifiers)

- Full-wave rectifiers are more commond to compared with half-wave rectifiers
- The utility supply is modeled as voltage source in series with internal impedance (primarily inductive)

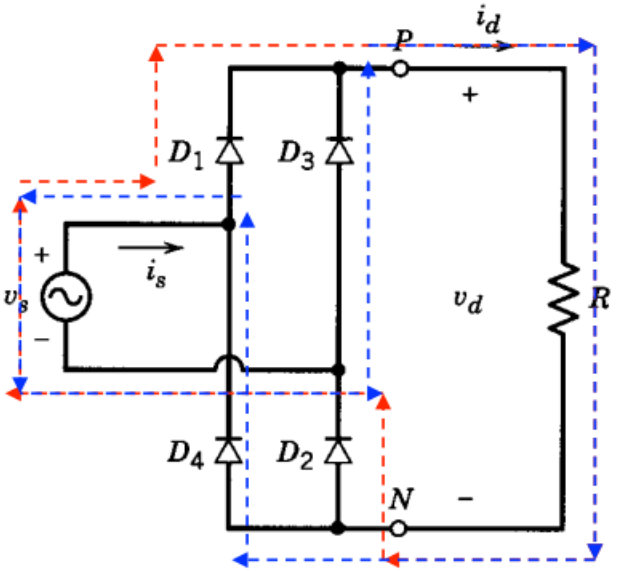


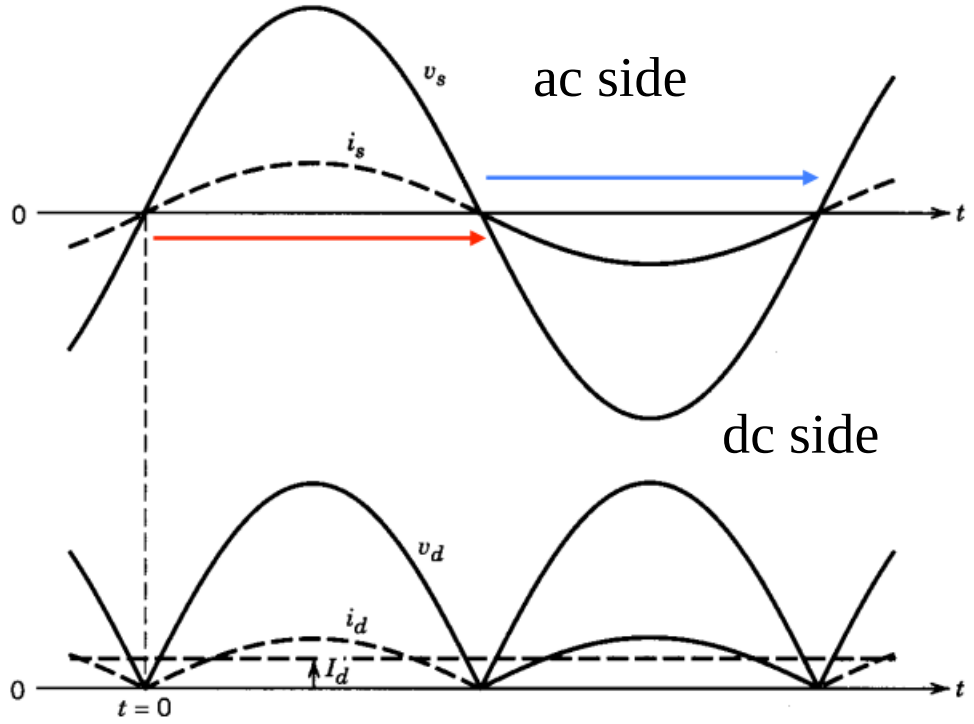
Conducting rules:

- Diode with highest anode potential is conducting
- Diode with lowest cathode potential is conducting

Single-phase diode bridge rectifiers (full-wave rectifiers)

Resistive load





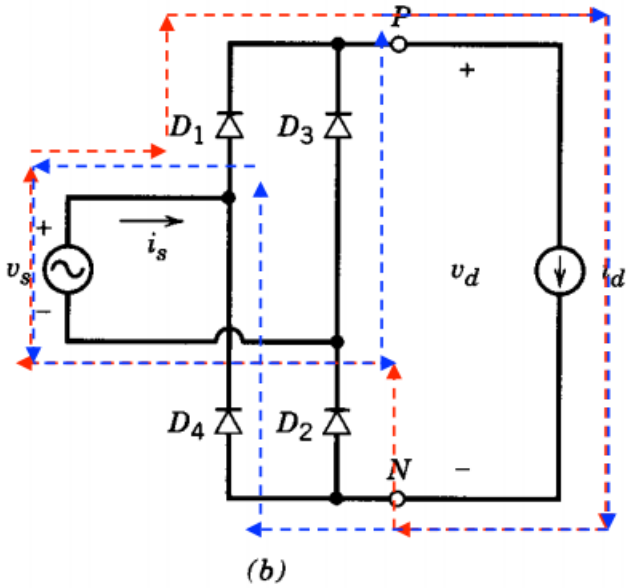
Average output voltage

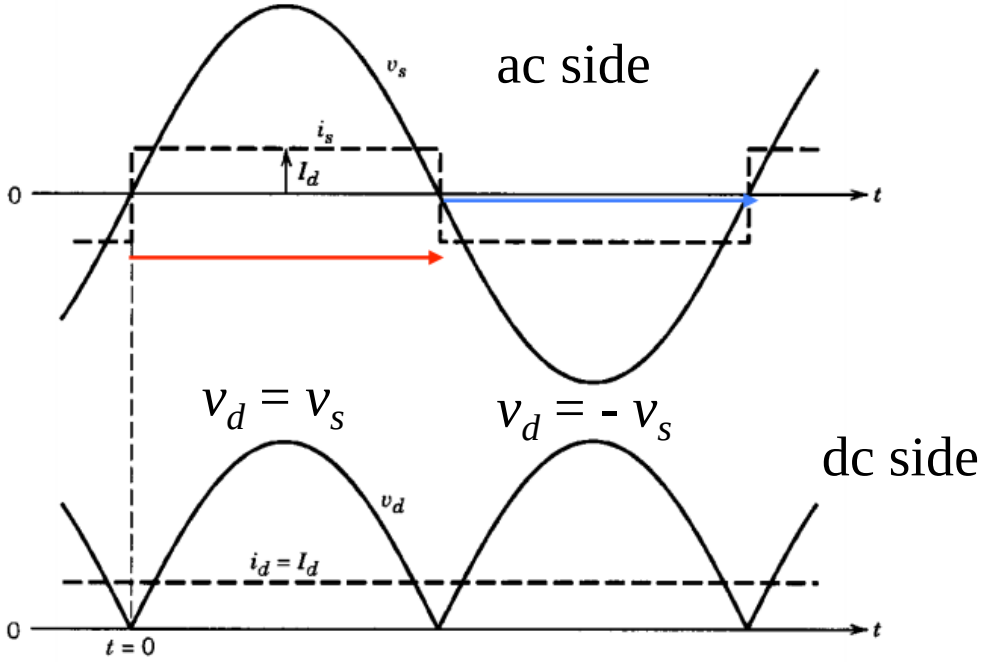
$$V_{d0} = \frac{1}{T/2}\int_0^{T/2}\sqrt{2}V_s \sin\omega t dt = \frac{2}{\pi}\sqrt{2}V_s = 0.9V_s$$

Single-phase diode bridge rectifiers (full-wave rectifiers)

Inductive load (large inductor)

I_d is the approximation to a situation where the large inductor is connected in series at dc output of the rectifier for filtering

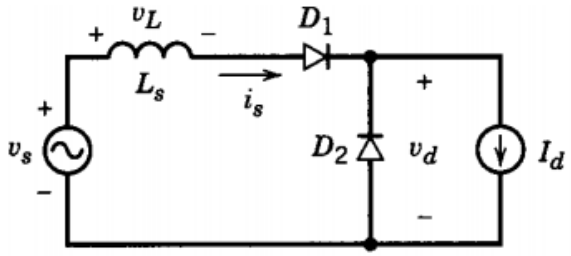


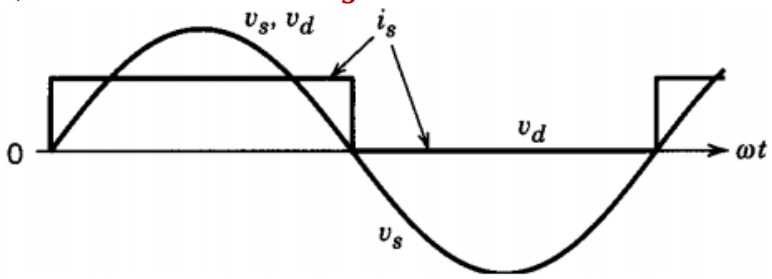


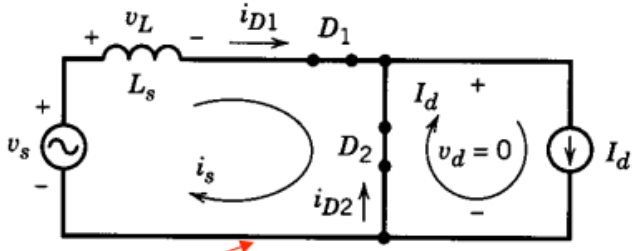
Note: if $L_s = 0$, is changes from I_d to $-I_d$ (and vice versa) instantaneously

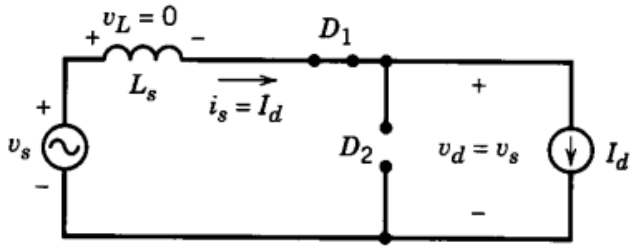
Single-phase diode bridge rectifiers (full-wave rectifiers)

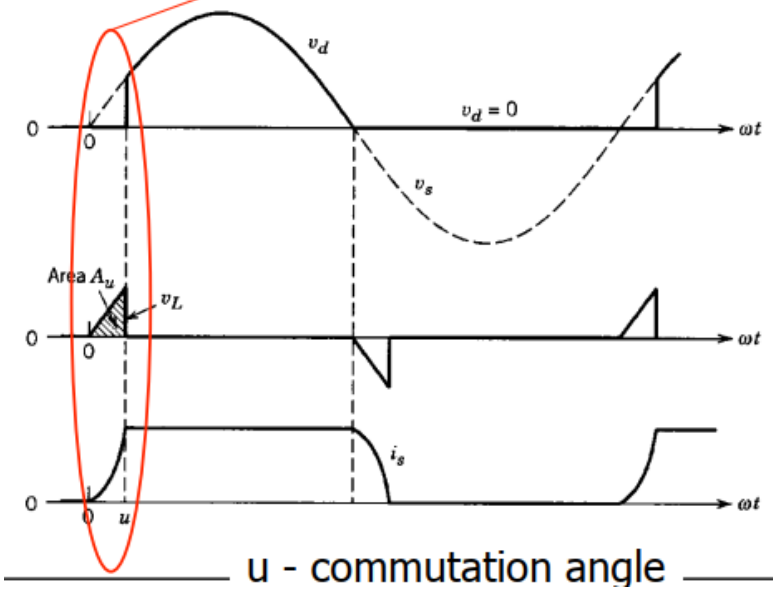
Inductive load (large inductor) – Effect of L_s











u - commutation angle

$$v_L = \sqrt{2}\ V_s\ \sin\omega t = L_s \frac{di_s}{dt} \qquad 0 < \omega t < u$$

$$i_s(t) = i_s(0) + \frac{1}{L_s}\int_0^t \sqrt{2}\ V_s\ \sin\omega t\ d(t)$$

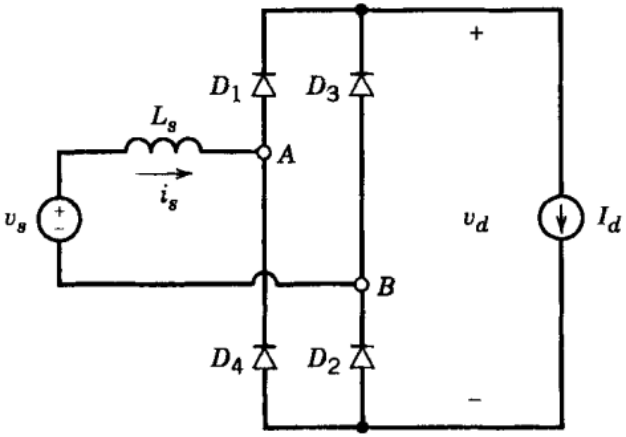
$$i_s(0) = 0 \qquad i_s(u/\omega) = I_d$$

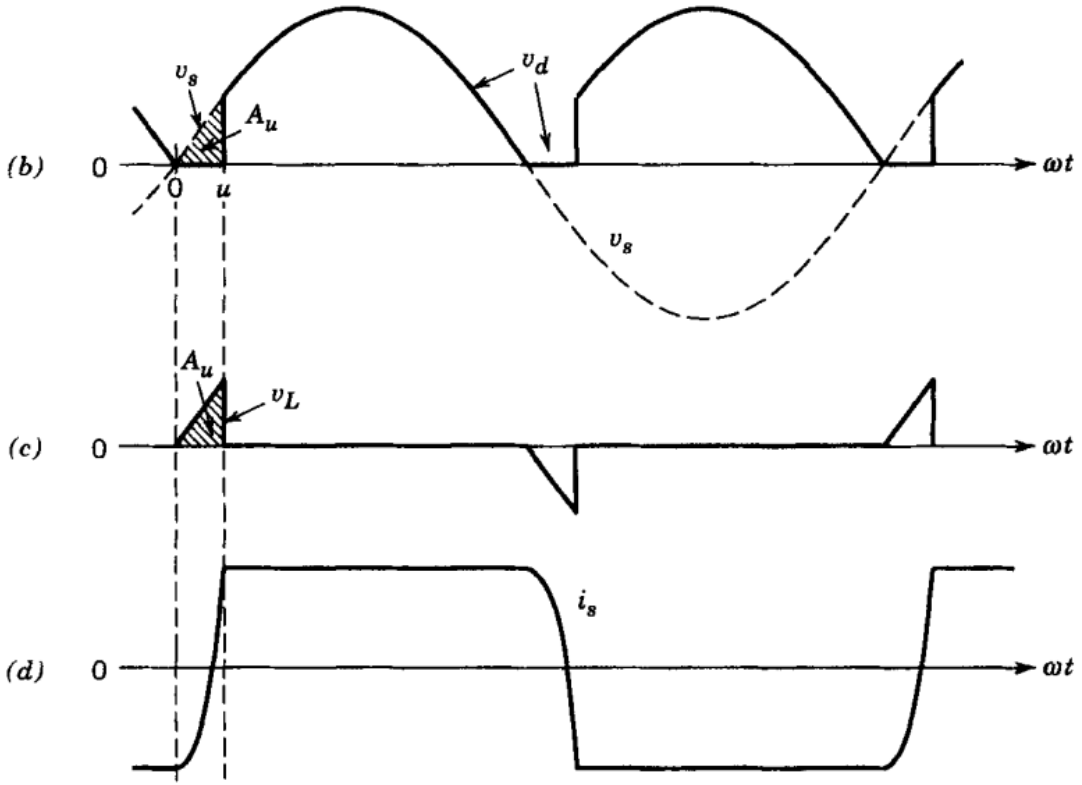
$$\omega L_s\ I_d = \sqrt{2}\ V_s\ (1 - \cos u)$$

$$\cos u = 1 - \frac{\omega L_s I_d}{\sqrt{2}\ V_s} \qquad A_u = \int_0^u \sqrt{2}\ V_s \sin\omega t\ d(\omega t)$$

Single-phase diode bridge rectifiers (full-wave rectifiers)

Inductive load (large inductor) – Effect of L_s

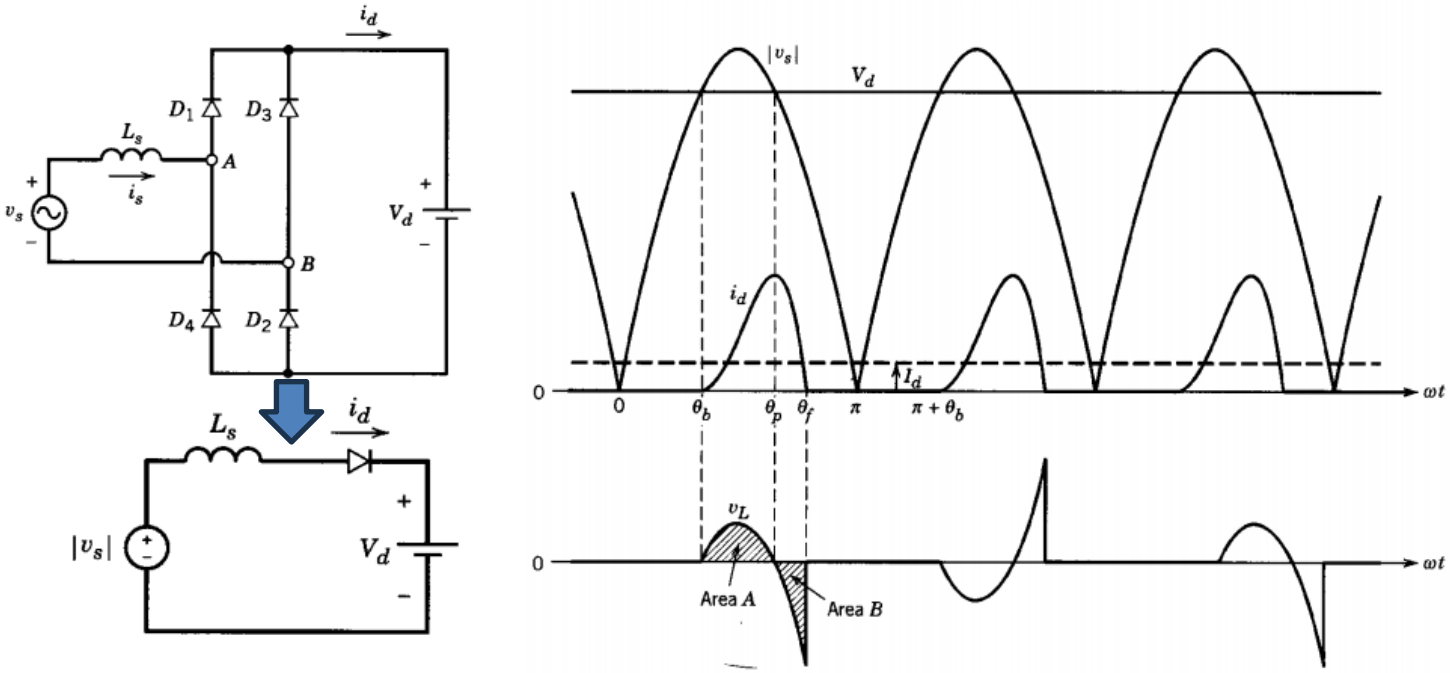




Single-phase diode bridge rectifiers (full-wave rectifiers)

Capacitive load (large capacitor)

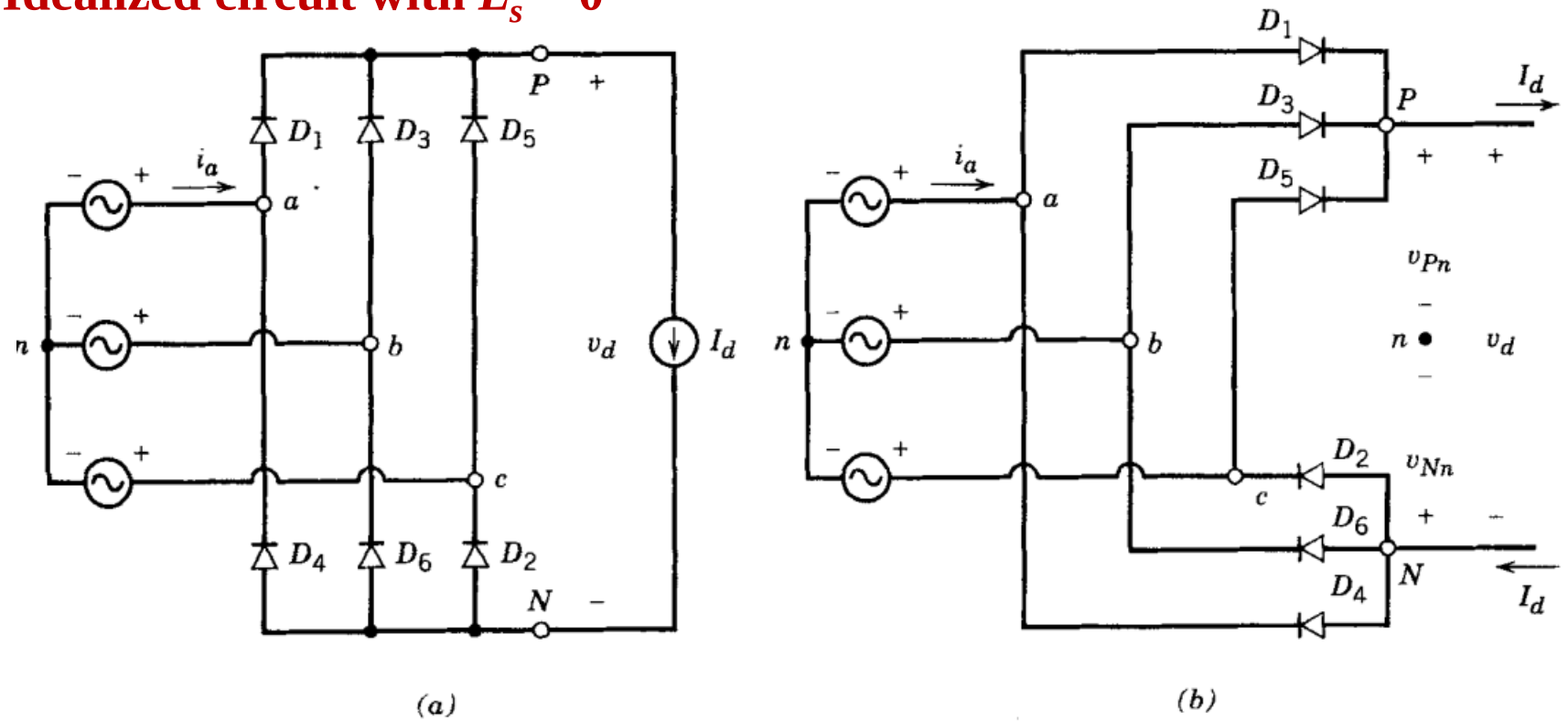
V_d is the approximation to a situation where the large capacitor is connected in series at dc output of the rectifier for filtering



- When v_s exceeds V_d at θ_b, D_1, D_2 start to conduct
- The current i_d reaches its peak at θ_p, beyond which v_L becomes negative
- The current becomes zero at θ_f when the volt-cross areas A and B become equal and negative of each other

Three-phase, full-bridge diode rectifiers

Idealized circuit with $L_s = 0$



Conducting rules:

- In the top group, the diode with its anode at the highest potential will conduct, the other two become reversed biased
- In the bottom group, the diode with its cathode at the lowest potential will conduct, the other two become reversed biased

Three-phase, full-bridge diode rectifiers

Idealized circuit with $L_s = 0$

$$v_d = v_{Pn} - v_{Nn}$$

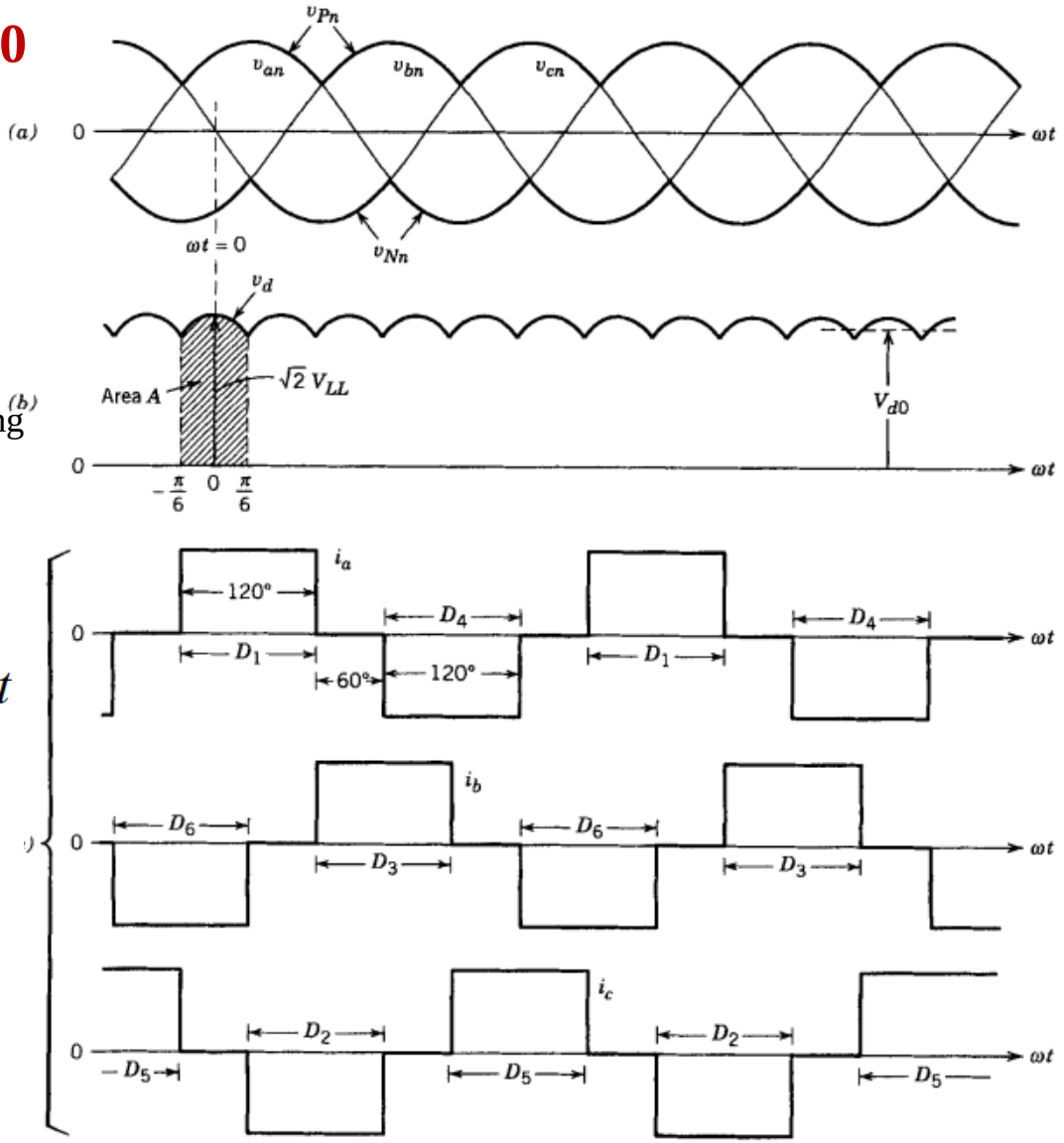
$$i_a = \begin{cases} I_d & \text{when D1 is conducting} \\ -I_d & \text{when D4 is conducting} \\ 0 & \text{when neither D1 or D4 is conducting} \end{cases}$$

For: $$-\frac{1}{6}\pi < \omega t < \frac{1}{6}\pi$$

$$v_d(t) = v_{ab}(t) = \sqrt{2}\ V_{LL}\ \cos \omega t$$

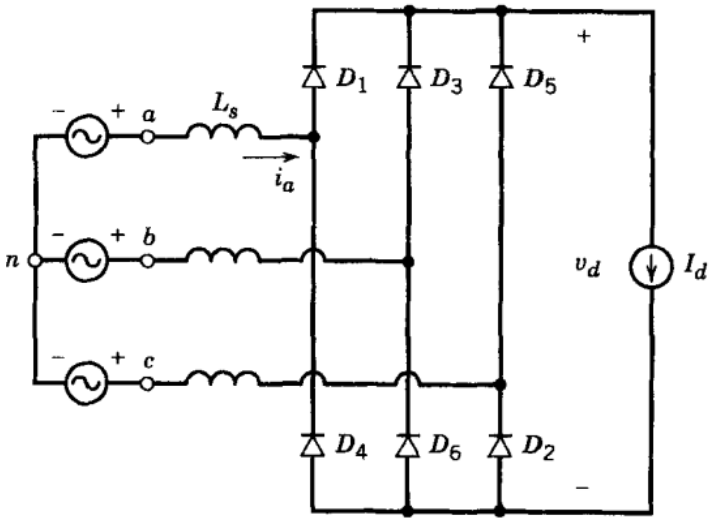
$$V_{do} = \frac{1}{\pi/3} \int_{-\pi/6}^{\pi/6} \sqrt{2}\, V_{LL} \cos \omega t$$

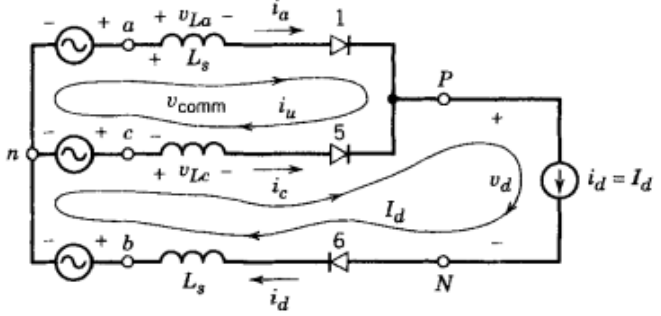
$$= \frac{3}{\pi} \sqrt{2}\, V_{LL} = 1.35\, V_{LL}$$



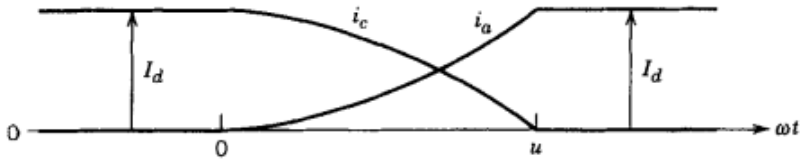
Three-phase, full-bridge diode rectifiers

Effect of L_s on current commutation

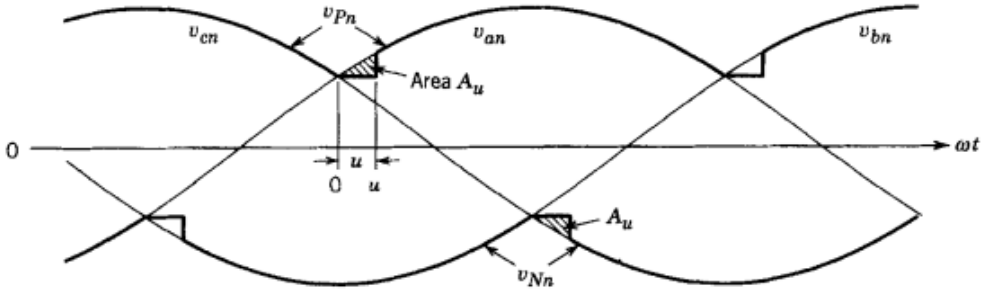




(a)



(b)



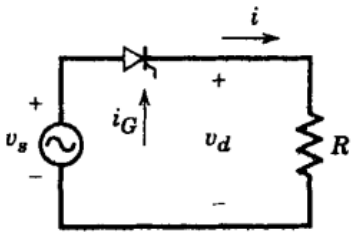
Part 2

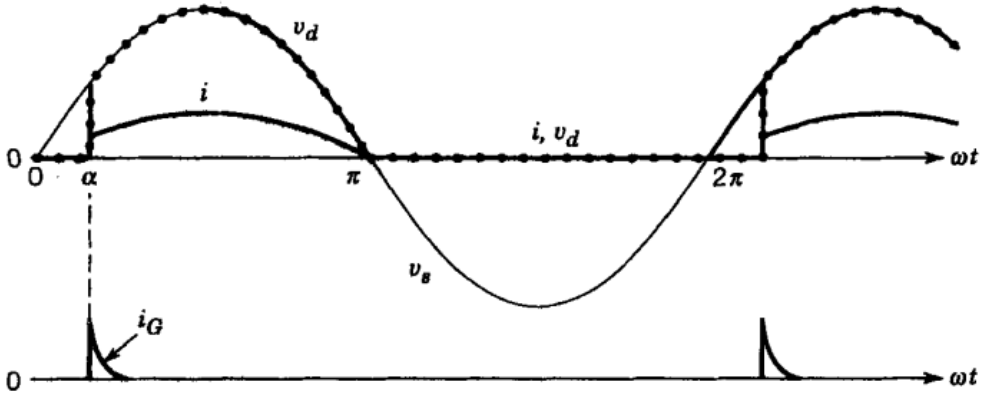
Thyristor Rectifiers (AC to Controlled DC)

Basic thyristor circuits

- Two conditions for thyristor to conduct: i) thyristor must be forward-biased, ii) a pulse current must be applied to the gate of thyristor

Resitive load





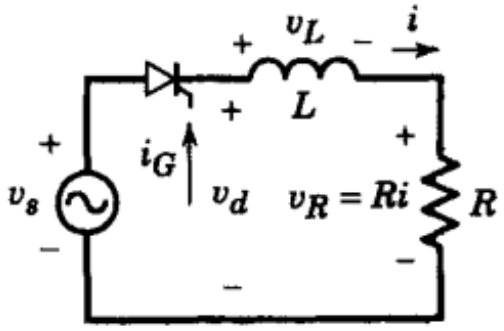
- When $v_s > 0$, the current is zero until a positive gate pulse is applied at $\omega t = \alpha$ (thyristor starts to conduct)
- The thysistor blocks the current when $v_s < 0$
- By adjusting α, the average value of the load voltage v_d can be controlled

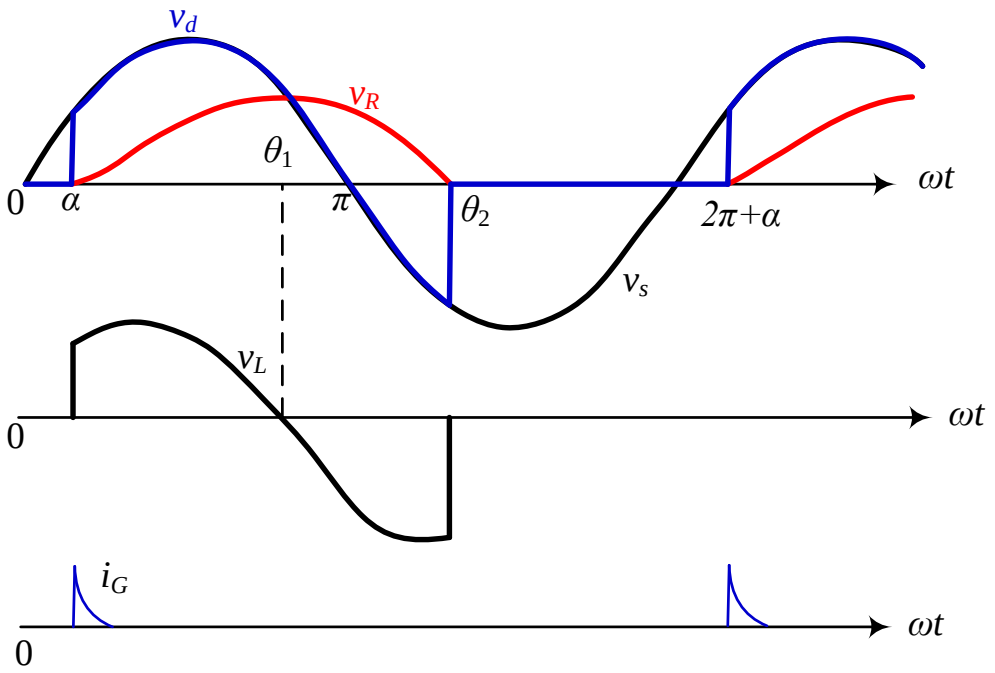
$$V_{d0} = \frac{1}{2\pi}\int_{\alpha}^{\pi}\sqrt{2}V_s \sin(\omega t)\,d(\omega t) = \frac{V_s}{\sqrt{2}\pi}(1+\cos\alpha)$$

Basic thyristor circuits

- Two conditions for thyristor to conduct: i) thyristor must be forward-biased, ii) a pulse current must be applied to the gate of thyristor

Inductive load



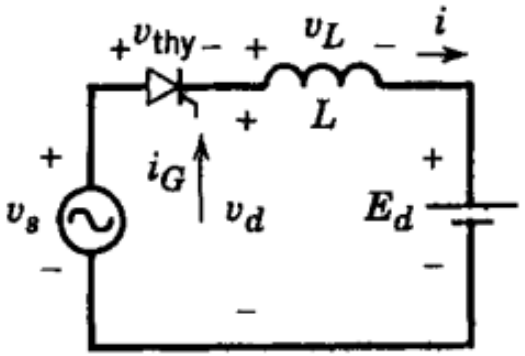


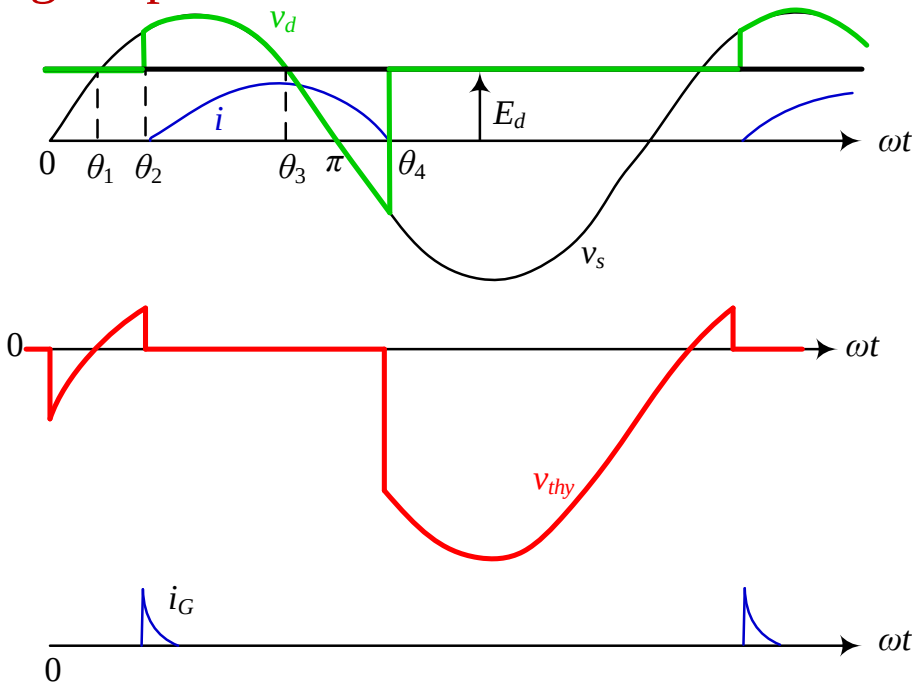
- Due to stored energy in the inductor, the current continuous to flow for a while even after v_s has become negative
- The voltage across inductor $v_L = L\dfrac{di}{dt} = v_s - v_R = v_s - Ri$

Basic thyristor circuits

- Two conditions for thyristor to conduct: i) thyristor must be forward-biased, ii) a pulse current must be applied to the gate of thyristor

Load with internal dc voltage/large capacitor

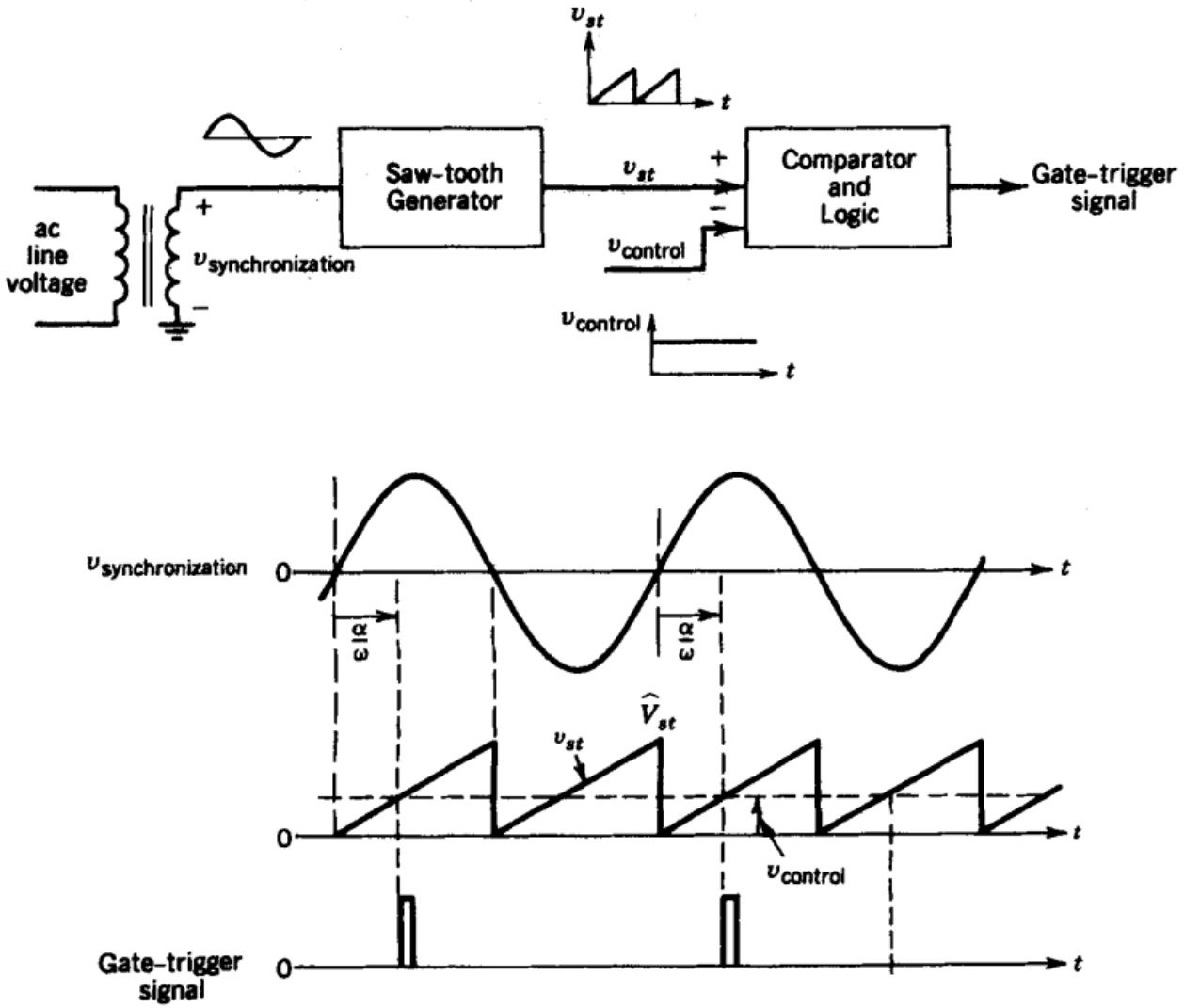




- The thyristor is forward-biased at $\omega t = \theta_1$ where $v_s > E_d$ and start to conduct at $\omega t = \theta_2$
- The voltage across inductor $v_L = L\dfrac{di}{dt} = v_s - E_d$

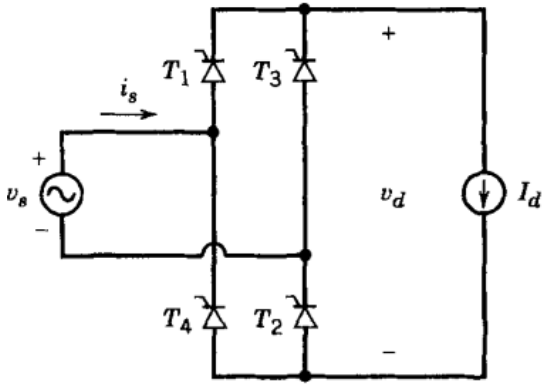
Basic thyristor circuits

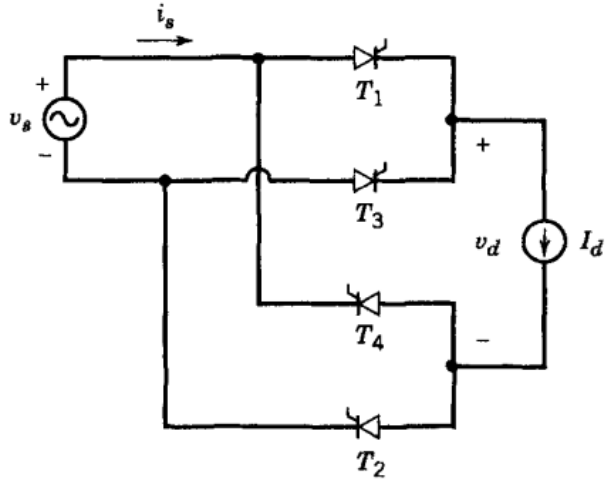
Thyristor gate trigging



Single phase thyristor bridge rectifier

Idealized circuit with $L_s = 0$ and $i_d(t) = I_d$

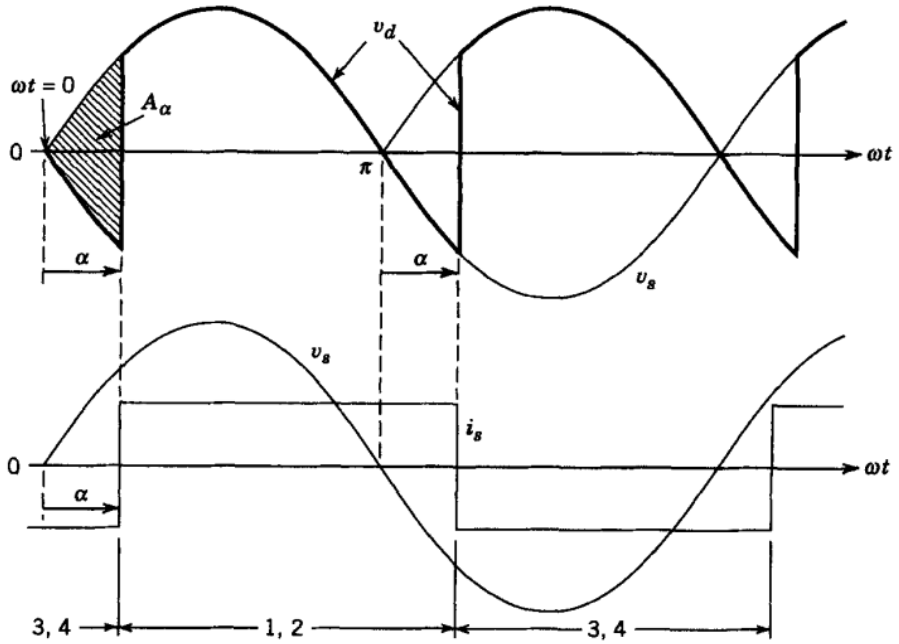




- The average output voltage

$$V_{d0} = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} \sqrt{2}V_s \sin(\omega t)\, d(\omega t)$$

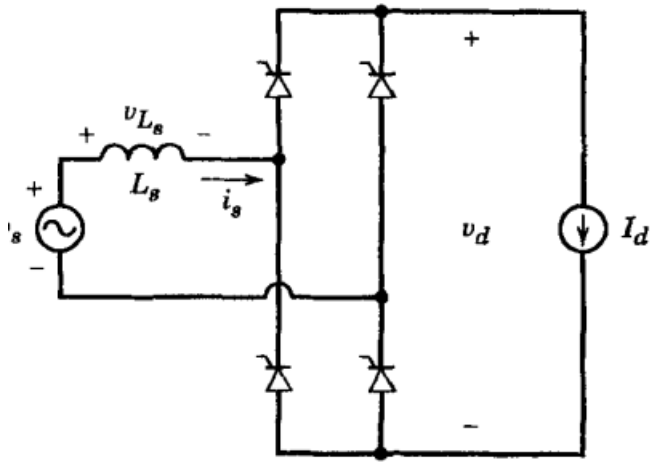
$$= \frac{2\sqrt{2}}{\pi} V_s \cos\alpha = 0.9 V_s \cos\alpha$$



Single phase thyristor bridge rectifier

Effect of L_s

- Since L_s # 0, the current is cannot change from $-I_d$ to I_d simultantaneously => the commutation time (u) is needed
- During commutation time, all four thyristors conduct, therefore $v_d = 0$.

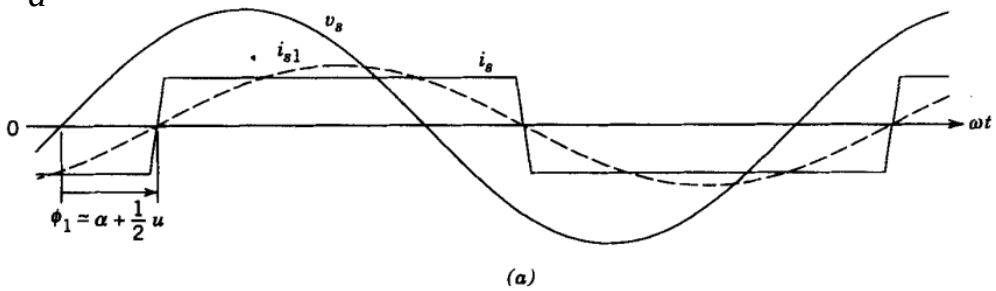


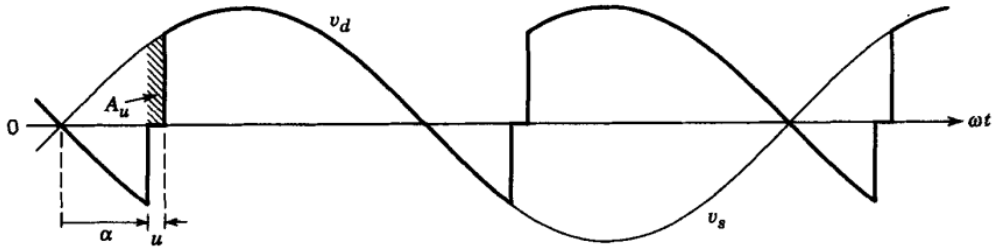
$$v_s = v_{Ls} = L_s \frac{di_s}{dt}$$

$$\int_{\alpha}^{\alpha+u} \sqrt{2}V_s \sin(\omega t)\, d(\omega t)$$

$$= \omega L_s \int_{-I_d}^{I_d} di_s = 2\omega L_s I_d$$

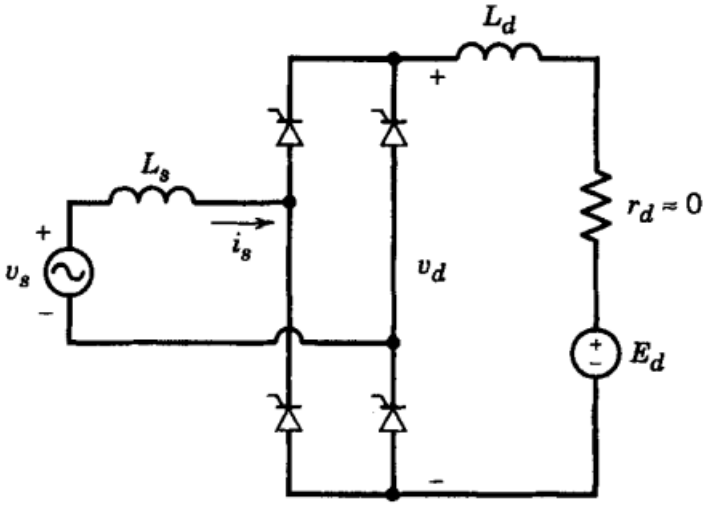
$$\Rightarrow \cos(\alpha + u) = \cos\alpha - \frac{2\omega L_s I_d}{\sqrt{2}V_s}$$





Single phase thyristor bridge rectifier

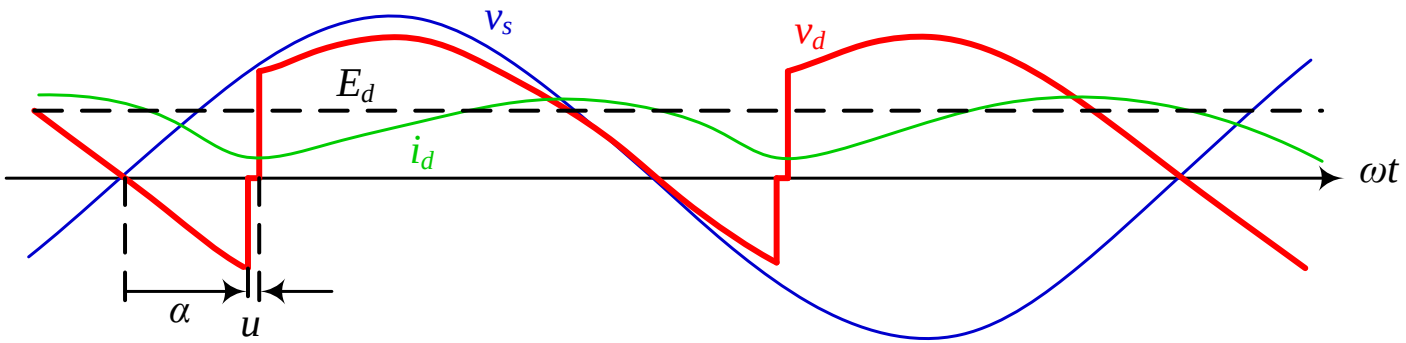
More practical rectifiers



- Due to L_s
- There is a finite commutation interval u
- v_d differs from v_s by the voltage drop across L_s
- The average output voltage

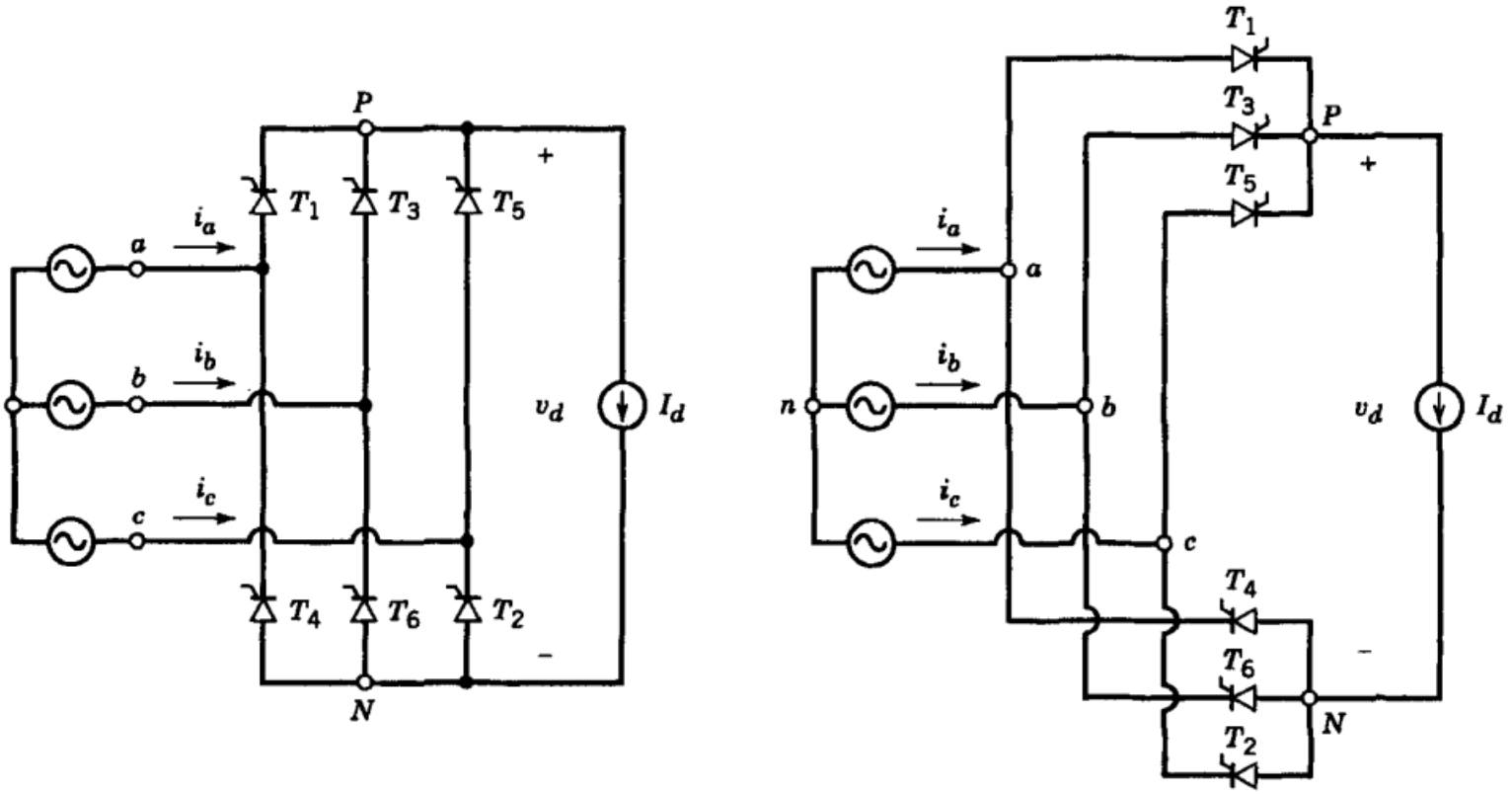
$$V_d \approx 0.9V_s \cos\alpha - \frac{2}{\pi}\omega L_s I_{d,\min}$$

- The average output current is determined by $V_d = r_d I_d + E_d$ (the average voltage across L_d is zero)



Three phase thyristor bridge rectifier

Idealized circuit with $L_s = 0$ and $i_d(t) = I_d$



- The current I_d flows through one of the thyristors of the top group (thyristors 1, 3, 5) and one of the bottom group (thyristors 2, 4, 6)

Three phase thyristor bridge rectifier

Idealized circuit with $L_s = 0$ and $i_d(t) = I_d$

- For $\alpha = 0$

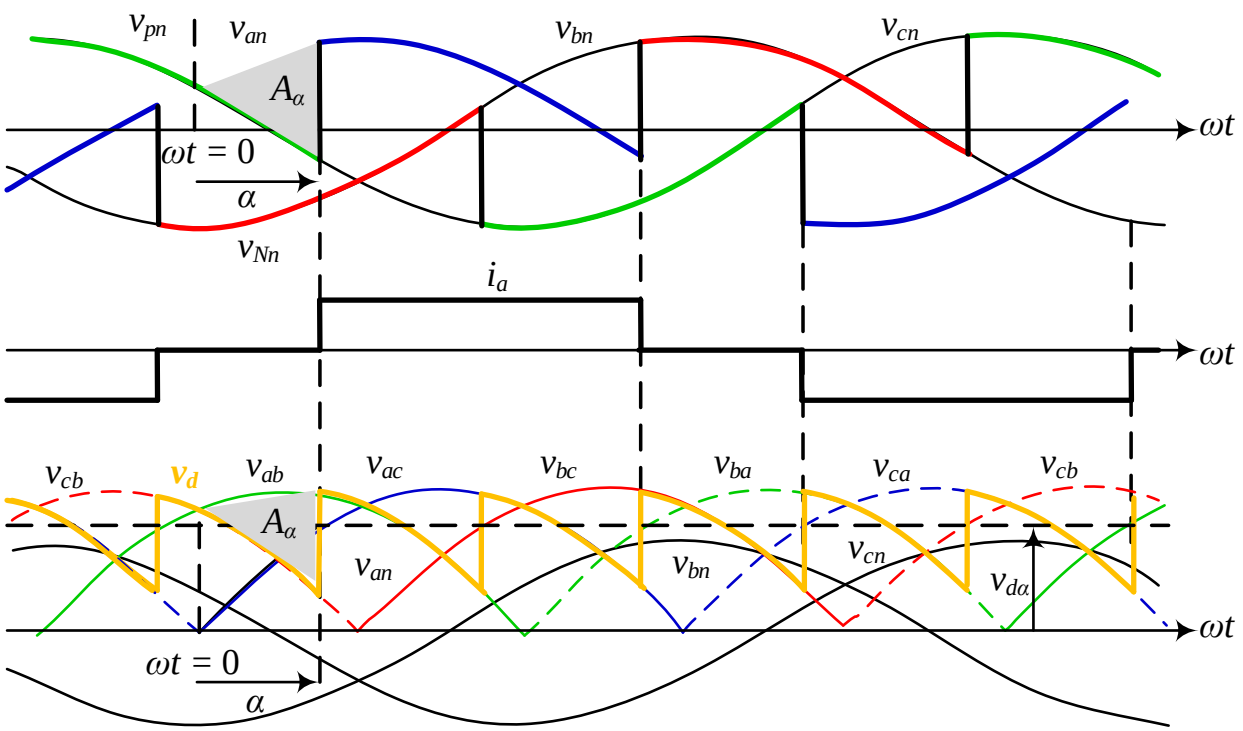
$$V_{d0} = \frac{3\sqrt{2}}{\pi} V_{LL} = 1.35 V_{LL}$$

(see diode rectifier)

- For $\alpha > 0$

(thyristor rectifier)

$$V_{d\alpha} = V_{d0} - \frac{A_\alpha}{\pi/3}$$

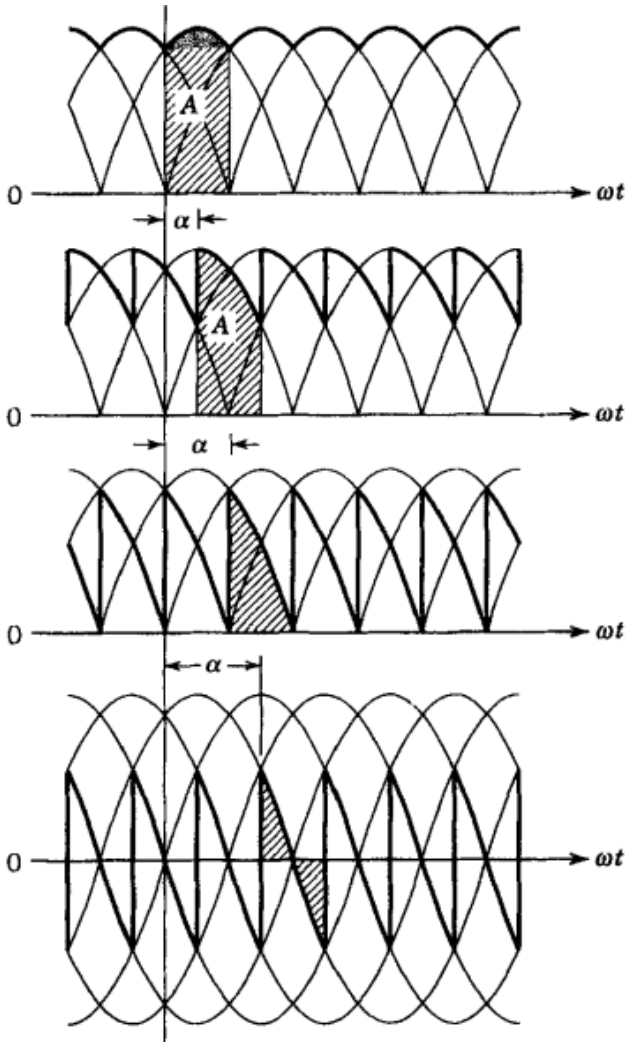


$$A_\alpha = \int_0^\alpha (v_{ab} - v_{bc})\, d\omega t = \int_0^\alpha v_{ac}\, d\omega t = \int_0^\alpha \sqrt{2} V_{LL} \sin(\omega t)\, d\omega t = \sqrt{2} V_{LL} (1 - \cos\alpha)$$

$$\Rightarrow V_{d\alpha} = \frac{3\sqrt{2}}{\pi} V_{LL} \cos\alpha = 1.35 V_{LL} \cos\alpha = V_{d0} \cos\alpha$$

Three phase thyristor bridge rectifier

Idealized circuit with $L_s = 0$ and $i_d(t) = I_d$

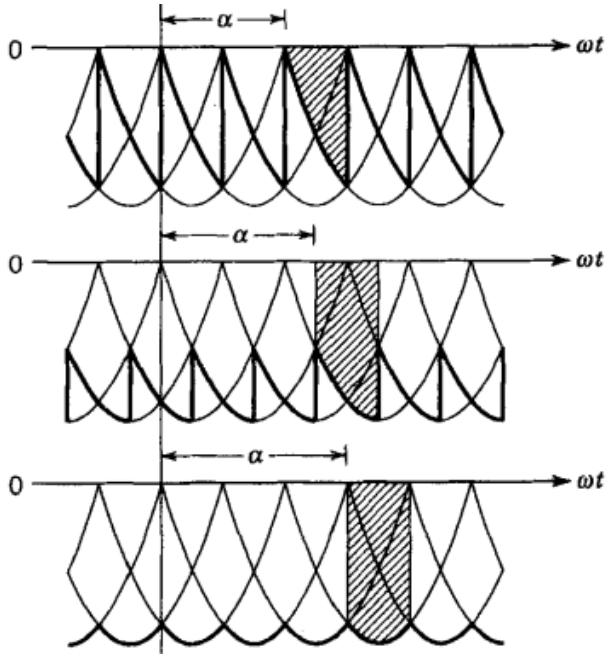


(a) $\alpha = 0$

(b) $\alpha = 30°$

(c) $\alpha = 60°$

(d) $\alpha = 90°$



(e) $\alpha = 120°$

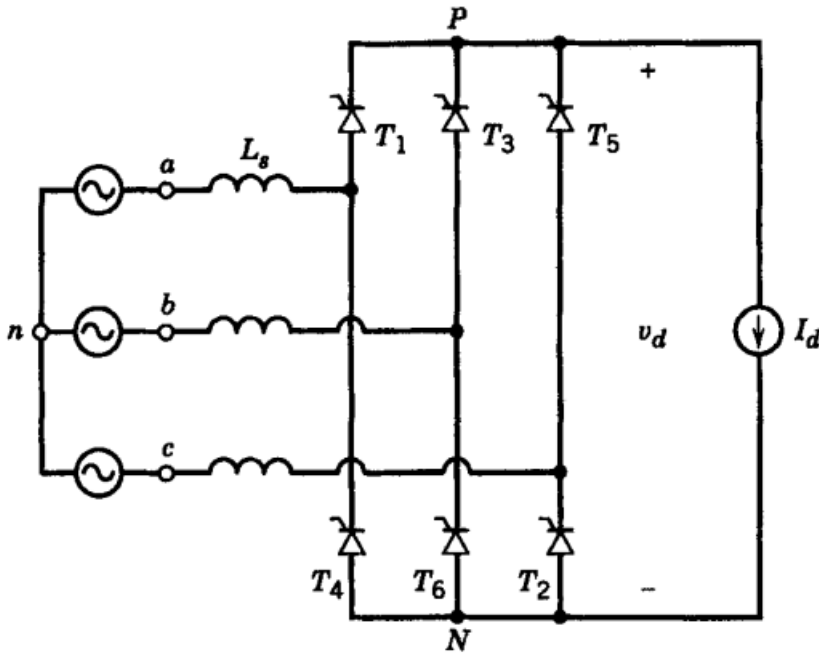
(f) $\alpha = 150°$

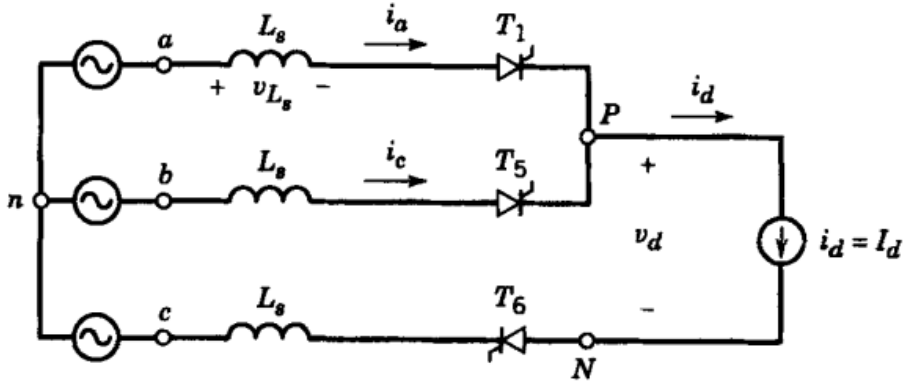
(g) $\alpha = 180°$

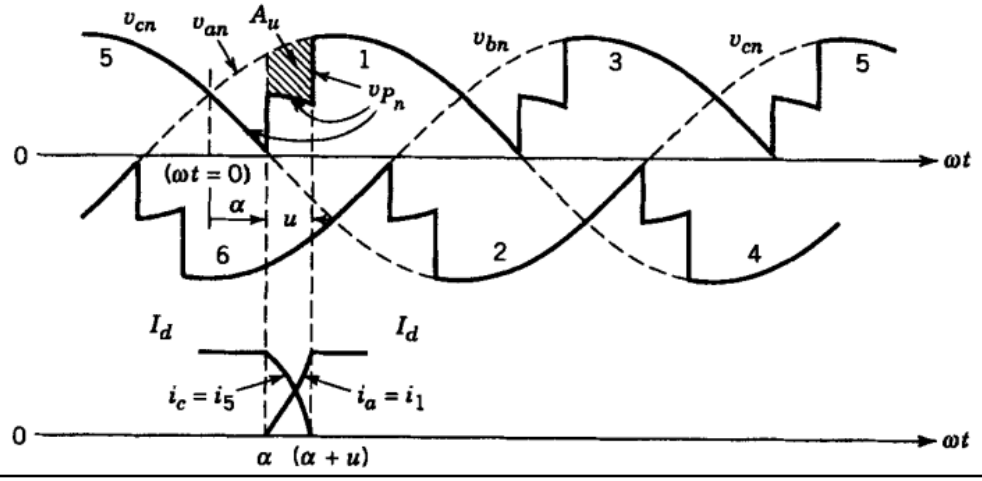
$$V_{d\alpha} = \frac{A}{\pi / 3}$$

Three phase thyristor bridge rectifier

Effect of L_s







Three phase thyristor bridge rectifier

Example

6-1 In the circuit of Fig. P6-1, v_{s1} and v_{s2} have an rms value of 120 V at 60 Hz, and the two are 180° out of phase. Assume $L_s = 5$ mH and $I_d = 10$ A is a dc current. For the following two values of the delay angle α, obtain v_{s1}, i_{s1}, and v_d waveforms. Calculate the average value V_d and the commutation interval u at (a) 45° and (b) 135°.

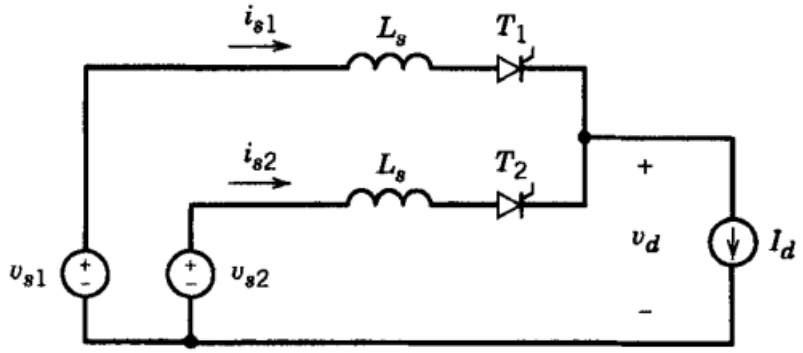


Figure P6-1